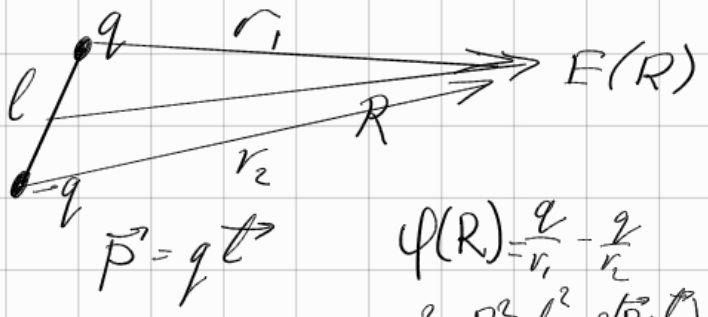


$\vec{p} = q\vec{\ell}$

$\varphi(R) = \frac{q}{r_1} - \frac{q}{r_2}$

$r_1^2 = R^2 + \frac{\ell^2}{4} - 2(\vec{R};\frac{\vec{\ell}}{2})$

$r_2^2 = R^2 + \frac{\ell^2}{4} + (\vec{R};\vec{\ell})$

$\nabla = \vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}$

$\vec{E} = \frac{q}{r^3}\vec{r}$

$\varphi = \frac{q}{r}$

$\vec{E} = -\vec{\nabla}\varphi = -q\left(i\frac{\partial r}{\partial x} + \dots\right)$

$r = \sqrt{x^2+y^2+z^2}$

$\frac{\partial r}{\partial x} = \frac{1}{2}\frac{2x}{r} = \frac{x}{r}$

$\vec{\nabla} r = \frac{\vec{r}}{r}$

$R >> r_1$

$r_1 = R\sqrt{1 + \frac{\ell^2}{4R^2} - \frac{(R\ell)}{R^2}}$

$\varphi = \frac{q}{R}\left[\left(1 + \frac{\ell^2}{4R^2} - \frac{(R;\ell)}{R^2}\right)^{\frac{1}{2}} - \left(1 + \frac{\ell^2}{4R^2} + \frac{(\ell;R)}{R^2}\right)^{-\frac{1}{2}}\right] \approx$

$\approx \frac{q}{R}\left[1 + \frac{R\ell}{2R^2} - 1 + \frac{R\ell}{2R^2}\right] = \frac{q(R;\ell)}{R^3}$

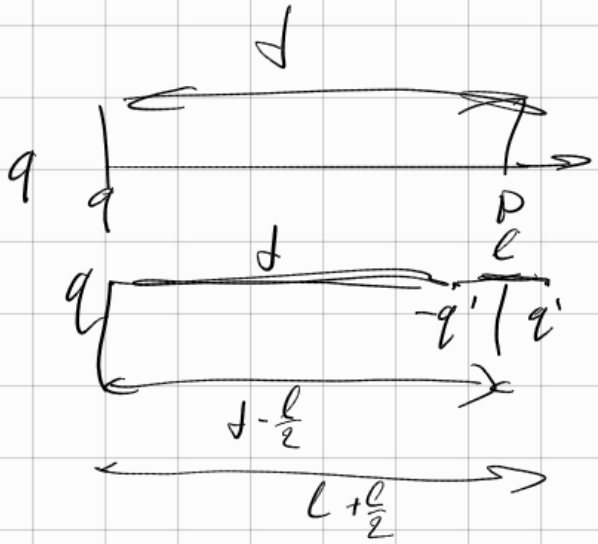
$-\vec{E} = \vec{\nabla}\varphi = q\vec{\nabla}\frac{(R;\ell)}{R^3} = q\left[-\frac{3}{R^4}(R;\ell)\frac{\vec{R}}{R} + \frac{1}{R^3}\vec{\nabla}(R;\ell)\right] =$

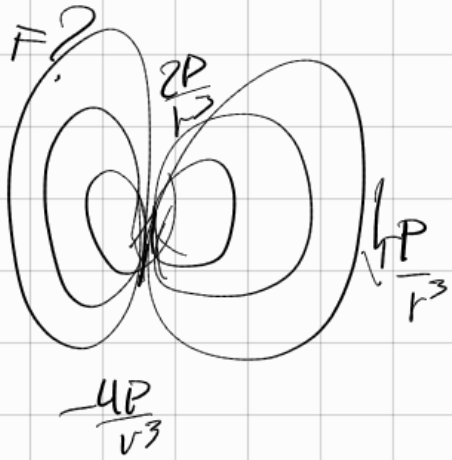
$\vec{\nabla}(R;\ell) = \vec{\nabla}(x\ell_x + y\ell_y + z\ell_z) = \ell_x i\frac{\partial x}{\partial x} + j\ell_y\frac{\partial y}{\partial y} + k\ell_z\frac{\partial z}{\partial z} = \vec{\ell}$

$= -\frac{3(\vec{P};\vec{R})\vec{R}}{R^5} + \frac{\vec{P}}{R^3}$

№ 1.7

$d;\ \vec{p}\parallel\vec{\ell}$





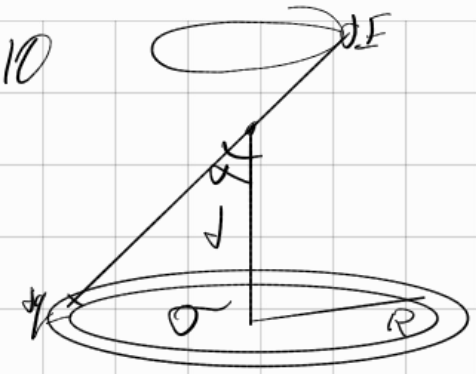
$E_1 = \frac{q}{(d-\frac{\ell}{2})^2}$

$E_2 = \frac{q}{(d+\frac{\ell}{2})^2}$

$F = -\frac{P}{\ell}\cdot\frac{q}{(d-\frac{\ell}{2})^2} + \frac{P}{\ell}\frac{q}{(d+\frac{\ell}{2})^2} = \frac{Pq}{\ell}\left[\frac{1}{(d+\frac{\ell}{2})^2} - \frac{1}{(d-\frac{\ell}{2})^2}\right] =$

$= \frac{Pq}{\ell d^2}\left[1 - \frac{\ell}{d} - \left(1 + \frac{\ell}{d}\right)\right] = \frac{Pq}{\ell d^2}\left(-2\frac{\ell}{d}\right) = -2\frac{Pq}{d^3} = (P;\nabla_P)E$

1.10



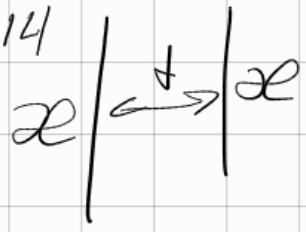
$$\sigma = \frac{Q}{\pi R^2}$$

$$dE = \frac{dq}{d^2+r^2} = \frac{\sigma 2\pi r\, dr}{d^2+r^2}$$

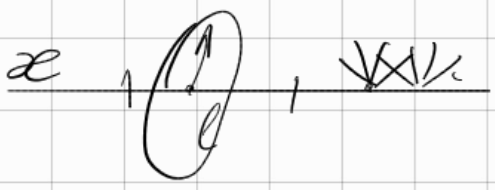
$$E(d) = E_{\parallel d} = \int_0^R 2\cos\alpha \frac{2\pi r\sigma dr}{d^2+r^2} = \int_0^R 2\frac{d\,2\pi r\sigma dr}{\sqrt{d^2+r^2}(d^2+r^2)} =$$

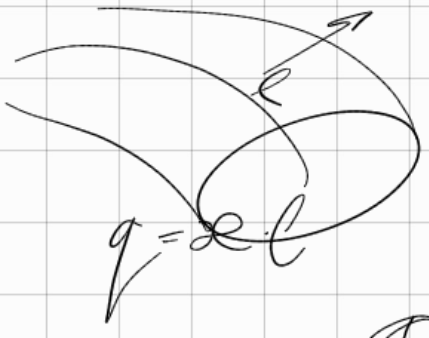
$$= 4\pi\sigma d \int_0^{R^2} \frac{dr^2}{(d^2+r^2)^{3/2}}$$

1.14



$$q(e) = \varkappa l$$

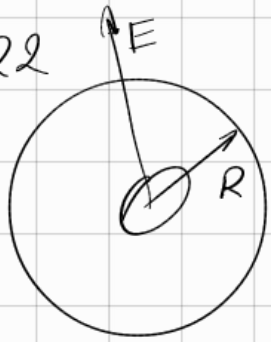




$$\Phi = 2\pi r l E = 4\pi \varkappa e l$$

$$E = \frac{2\varkappa e}{r}$$

1.22



$$\vec{E} = \frac{q\vec{R}}{R^3}$$



$$E(\rho_+) = \frac{\rho_+ \frac{4}{3}\pi r_1^3}{r^3}\vec{r_1} = \frac{4}{3}\pi\rho_+\vec{r_1}$$

$$E(\rho_-) = \frac{\rho_- \frac{4}{3}\pi r_1^3}{r_1^3}\vec{r_1} = \frac{4}{3}\pi\rho_-\vec{r_1}$$

$$\Rightarrow \frac{4}{3}\pi\rho_+(\vec{r_1} - \vec{r_1'})$$

1.23



$\sigma(\theta)$?



$E_n \cdot \oint = 4\pi\sigma \oint$

$E_n = E_0 \cos\theta \sim \sigma / R$

$\sigma(\theta) = C \cdot \cos\theta$

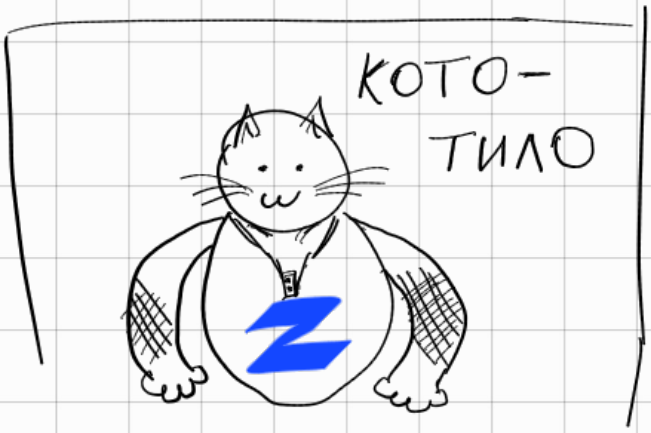
$$E(\theta) = 2\int_0^{\pi} \frac{\sigma(\theta) 2\pi R \sin\theta R \cos\theta d\theta}{R^2}$$

$$= 4\pi C \int_0^{\pi} \cos^2\theta \, d\cos\theta =$$

$$= \frac{4\pi}{3}(-C)\cos^3\theta \Big|_0^{\pi} =$$

$$= \frac{4}{3}\pi C \cdot 2 = E_0 \Rightarrow$$

$$\Rightarrow C = \frac{3}{8\pi} E_0$$



11.09. Потенциал

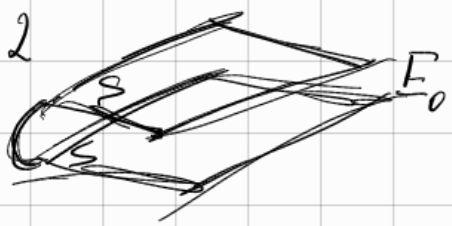
°2.1

$$E^2 = m^2c^4 + p^2c^2 = (mc^2 + E_k)^2$$

$$P_{He} = P_{Au}$$

$$U = \frac{q_{He} q_{Au}}{r_{min}}$$

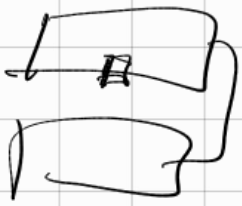
°2.2



$S = 1\ м^2$

$E_0 = 130\ В/м$

$q = ?$

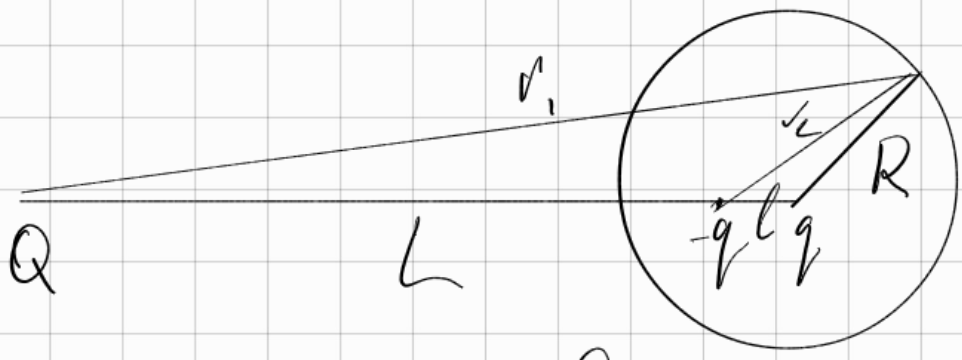


$$\oint \vec{E} d\vec{S} = 4\pi Q$$

$$ES = 4\pi Q = 4\pi\sigma S_0 = ES_0$$

$$q = \sigma S = \frac{ES}{4\pi}$$

°2.3



$$F = \frac{Qq}{(d-l)^2} + \frac{Q(-q)}{d^2} =$$

$$= \frac{Q^2 R d}{(d^2 - R^2)^2} + \frac{Q^2 R}{d^3}$$

$\varphi_S = const$

$$\varphi_S = \frac{Q}{r_1} + \frac{q}{r} - \frac{q}{r_2}$$

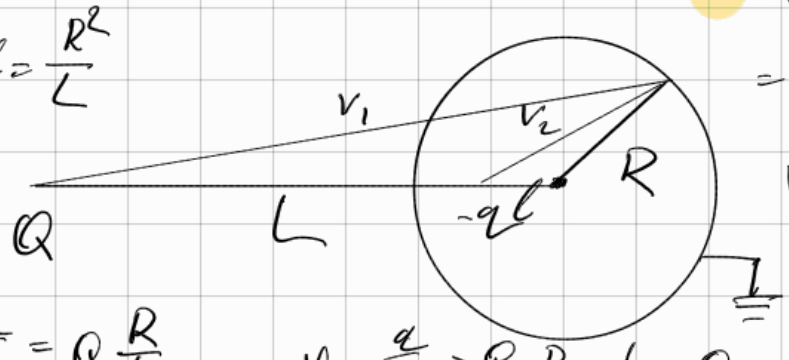
$$F = \frac{Qq}{(d-l)^2} = -\frac{Q^2 R}{d\left(d - \frac{R^2}{d}\right)^2} =$$

$$= \frac{-Q^2 R d}{(d^2 - R^2)^2}$$

$$\varphi_S = 0$$

$$\frac{q}{Q} = \frac{r_2}{r_1} = \frac{R}{l} \Rightarrow l = \frac{R^2}{L}$$

$$\varphi_S = 0 = \frac{Q}{r_1} = \frac{q}{r_2} \Rightarrow$$



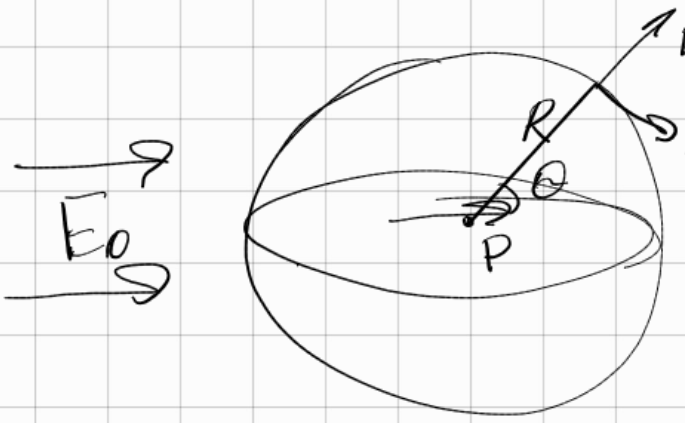
$$\Rightarrow q = Q\frac{r_2}{r_1} = Q\frac{R}{L}$$

$$\varphi_S = \frac{q}{R} = Q\frac{R}{L}\cdot\frac{1}{R} = \frac{Q}{L}$$

1.25

$$E_g = \frac{3(\vec{p}\vec{r})\vec{r}}{r^5} - \frac{\vec{p}}{r^3}$$

$$E_{out} = \vec{E}_0 + \vec{E}_g$$



$$E_{n\,out} = E_0\cos\theta + \frac{3P}{R^3}\cos\theta - \frac{P\cos\theta}{R^3} = \cos\theta = E_0 + \frac{2P}{R^3} = 4\pi\sigma(\theta)$$

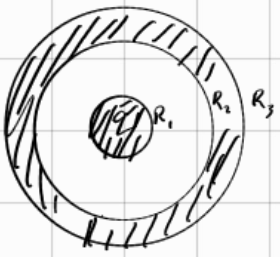
$$E_{\tau\,out} = E_0\sin\theta - \frac{P\sin\theta}{R^3} = 0 \qquad P = E_0 R^3$$

$$4\pi\sigma(\theta) = \cos\theta\left(E_0 + \frac{2}{R^3}\cdot E_0 R^3\right) = 3E_0\cos\theta$$

$$\sigma = \frac{3}{4\pi}E_0\cos\theta$$

$$E_{out} = \vec{E}_0' + \frac{3(\vec{E}_0 R^3;\vec{R})\vec{R}}{R^5} - \frac{\vec{E}_0 R^3}{R^3} = 3E_0\cos\theta \quad \begin{cases} E_0 \Rightarrow 3\cos\theta = 1 \\ 2E_0 \Rightarrow 3\cos\theta = 2 \end{cases}$$

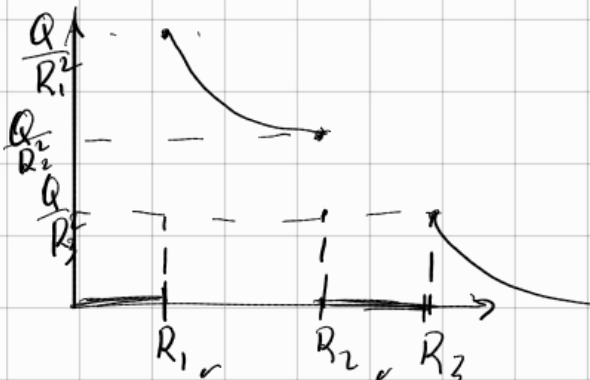
2.3

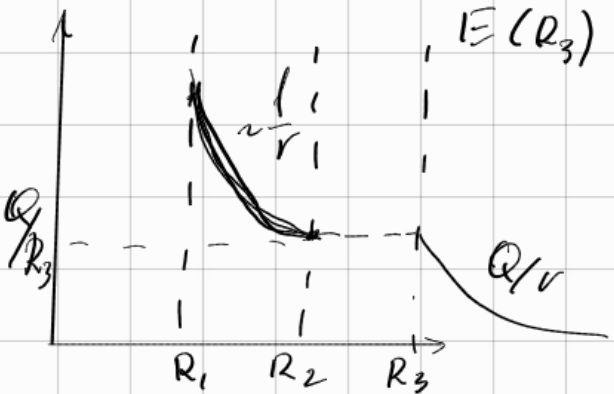


$$\oint E_n\,dS = 4\pi Q$$

$$E(R_3)\cdot 4\pi R^2 = 4\pi Q$$

$$E(R_3) = \frac{Q}{R_3^2}$$





$$\varphi = -\int_{\infty}^{r} E\,dr = -\int_{\infty}^{r}\frac{Q}{r^2}dr = \frac{Q}{r}$$

$$\varphi(r) = \frac{Q}{r} + C \qquad C = Q\left(\frac{1}{R_3} - \frac{1}{R_2}\right)$$

$$\varphi(R_2) = \frac{Q}{R_2} + C = \frac{Q}{R_3}$$